METRIC
MIL-DTL-24441/21B(SH)
27 August 2009
SUPERSEDING
MIL-DTL-24441/21A(SH)
19 May 1999

DETAIL SPECIFICATION SHEET

PAINT, EPOXY-POLYAMIDE, HAZE GRAY FORMULA 151, TYPE III

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24441.

FORMULA: This formula covers haze gray epoxy-polyamide paint designated Navy Formula 151, Type III for interior or exterior use. The paint shall consist of the ingredients specified in the quantities specified.

Component A[12/]

Ingredients	Kilograms	Pounds
Thixatrope[1/]	---	---
Polyamide[2/]	9.53	21.0
Polyamide adduct[3/]	134.72	297.0
Magnesium silicate[4/]	180.53	398.0
Titanium dioxide[5/]	48.08	106.0
Black iron oxide[6/]	7.26	16.0
Butyl alcohol[7/]	118.39	261.0

Component B[12/]

Ingredients	Kilograms	Pounds
Thixatrope[8/]	---	---
Epoxy resin[9/]	240.41	530.0
Magnesium silicate[4/]	120.2	265.0
Naphtha[10/]	100.25	221.0

See footnotes on next page.

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QUANTITATIVE REQUIREMENTS: The paint shall meet the following quantitative requirements and the qualitative requirements of section 3 of the general specification. The components A and B shall be mixed 1:1 by volume for mixed component tests. Tests shall be performed in accordance with the general specification.

Requirements	Component A		Component B		Mixed components	
	Min	Max	Min	Max	Min	Max
Pigment content, percent (%)	46.0	50.0	24.8	28.8	---	---
Volatiles, %	28.1	32.1	19.6	23.6	---	---
Nonvolatile vehicle, %	19.9	23.9	49.6	53.6	---	---
Water, %	---	1.5	---	0.5	---	---
Coarse particles, %	---	0.3	---	0.3	---	---
Consistency, grams	---	460	---	275	---	---
Kilograms per liter (kg/L) (pounds per gallon (lb/gal))	1.31 (10.9)	1.35 (11.3)	1.20 (10.0)	1.32 (11.0)	---	---
Set to touch, hours						
a. at 4.4 °C (40 °F)	---	---	---	---	---	2
b. at 23 °C (73 °F)	---	---	---	---	---	2
Dry-hard, hours						
a. at 4.4 °C (40 °F)	---	---	---	---	---	24
b. at 23 °C (73 °F)	---	---	---	---	---	6
Fineness of grind, NS	5	---	5	---	---	---
Flash point, °C (°F)	35.6 (96)	---	37.8 (100)	---	35.6 (96)	---
Titanium dioxide, % of pigment	20	---	---	---	---	---
Pot life, hours at 23 °C (73 °F)	---	---		---	5	---
Sag resistance, micrometers (mils)	---	---	---	---	225 (9)	---
Color of dry film to approximate the standard color chip	---	---	---	---	Conform	
Weight per epoxy, vehicle	---	---	175	195	---	---
Contrast ratio, 3-mil dry film thickness	---	---	---	---	0.98	---
Gloss, 60 degree specular, %	---	---	---	---	---	30
VOC, grams per liter (g/L) (lb/gal)	---	---	---	---	---	340 (2.8)

1/ Thixatrope to be used is the manufacturer's choice. In the development of component A, 6.80 kilograms (15 pounds) (7.95 liters (2.1 gallons)) of Disparlon NS-30 made by King Industries were used. Manufacturer is responsible for choosing a thixatrope that meets all the requirements herein, including shelf life. Thixatrope is a pigment for calculation purposes.

2/ GENAMID 2000, Cognis Corp.; ANCAMIDE 507, Air Products and Chemical Corp.; EPOTUF SF7791, REICHHOLD Chemical.

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3/ VERSAMID 280B75, Cognis Corp.; ANCAMIDE 700B75, Air Products and Chemical Corp.; EPOTUF SF7792, REICHHOLD Chemical.

4/ Nicron 503, Luzenac, Inc.

5/ Titanium dioxide conforming to ASTM D476, Type IV. In the development of component A, DuPont Tipure R960 was used.

6/ Black Iron Oxide Bk-5599, Elementis Pigments; Bayferrox Iron Oxide Black 318M, Lanxess Pigments

7/ Butyl alcohol conforming to ASTM D304.

8/ Thixatrope to be used is the manufacturer's choice. In the development of component B, 6.80 kilograms (15 pounds) (6.81 liters (1.8 gallons)) of Disparlon 6500 made by King Industries were used. Manufacturer is responsible for choosing a thixatrope that meets all the requirements herein, including shelf life. Thixatrope is a pigment for calculation purposes.

9/ EPON 828, HexionChemical Co.; ARALDITE 6010, Huntsman Advanced Materials; DER 331, Dow Chemical Corp.; EPO-TUF 37-140, REICHHOLD Chemical Corp.

10/ Conforming to ASTM D3734, Type I. In the development of component B, AMSCO Super High Flash Naphtha was used.

11/ Use FED-STD-595 color chip number 26270. Color shall approximately match the color chip.

12/ For VOC calculations, component A makes approximately 379 liters (100 gallons) and component B makes approximately 386 liters (102 gallons).

Use of alternate ingredients in this formula must have prior approval of the Naval Sea Systems Command. Approval will be based on review of data demonstrating equivalent physical, chemical, and performance characteristics of paint manufactured with the proposed alternate material and the requirements as specified in 3.4.1 of MIL-DTL-24441. Paint incorporating the proposed alternate ingredient shall conform to all the requirements of this military specification sheet and the general specification.

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity
Navy – SH
(Project 8010-2009-005)

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